Finding Similar Patient Records with TF-IDF and BM25

HI 744 Programming Assignment

Name: Krishna sai saketh Muddineni

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Project Goal and Dataset

This project aimed at constructing a system capable of identifying patient records similar to a particular patient's record, since searching for similar cases could help clinicians to find other patient experiences that are worthwhile reviewing. I carried out a comparison between two text-retrieval methods: TF-IDF and BM25.

The data folder had 8,579 patient records, each one stored in JSON file format. I used all of the records and did not select a sample. After preprocessing the case descriptions had an average length of 286.8 words. Each record also had a similar_patients field, and I used this as the reference when assessing the search results.

Methods

Prior to comparing the records, I processed the case descriptions. I changed the text to lowercase, got rid of the punctuation and any characters that are not letters, eliminated the common English stop words, and used the lemmatizer from NLTK's WordNet. These steps serve to reduce the differences between words and remove common words which are unlikely to show that one case is different from another. A result of this is that the numbers have been removed. This could be a limitation since ages, measurements, and dosages can have clinical significance.

For the baseline search method I used the TfidfVectorizer from scikit-learn. TF-IDF assigns greater weight to the words which help to distinguish one record from the others. I turned on smoothed inverse document frequency in order to maintain stable weights, used sublinear term frequency to reduce the impact of words that are repeated many times, and applied L2 normalization so that records of different lengths could be compared using cosine similarity.

I too made use of BM25, a method which ranks records by means of word frequency and document length; for this approach I set k1 to 1.5 and b to 0.75. The value of k1 determines the rate at which the advantage of repeated words levels off, while the value of b determines the extent to which document length affects the score.

For every patient, both methods ranked the other records and selected the five highest-scoring ones. I left out the patient's own record from the results.

Evaluation and Results

I gave the same consideration to similar-patient relevance scores of 1 and 2. Precision@5 is calculated by dividing the number of the five retrieved records that

were deemed relevant by five. Recall@5 is obtained by dividing the number of relevant records that appeared in the top five by the total number of relevant records for that patient.

For 3,777 records there were no similar patients listed; after following the assignment instructions I excluded those records from the recall averages. The precision was averaged over all 8,579 records. The results were:

Method	Average precision@5	Average recall@5
TF-IDF	0.0066	0.0289
BM25	0.0057	0.0255

TF-IDF obtained a slightly higher score than BM25 on both criteria. Its average precision@5 was 0.66% and its average recall@5 was 2.89%. For BM25, the average precision@5 was 0.57% and the average recall@5 was 2.55%.

The methods had 3.13 patients on average in their top-five results; this meant that they usually included a number of the same patients, even though each method also returned some different records.

Discussion and Limitations

The figures indicate that both methods identified only a small proportion of the records which had been marked as similar in the dataset; TF-IDF performed only slightly better in this instance, the difference between the two methods being minor. Furthermore, the top-five overlap shows that the methods generally made the same choices.

The results must be regarded as indicating agreement with the labels of the dataset; although a record which had not been marked as similar might still be of use to a clinician, it would not be counted as a match in this evaluation. Furthermore, patients whose records had no similar entries labeled could not be incorporated into the recall averages.

Preprocessing also has its limitations; removing numbers might result in the loss of useful clinical information and basic word cleaning is not capable of capturing the meaning of medical phrases or the relationships between symptoms. A later version could include the preservation of measurements and make use of medical terminology tools, and it would be useful to have a clinician examine some of the retrieved cases.

The project involved producing and comparing two methods for searching patient records. In this instance, TF-IDF achieved a slightly higher level of precision and recall, and on average the TF-IDF and BM25 methods had about three of their top five results in common. Although these methods offer a useful starting point for identifying similar records, one should bear in mind the low scores and the limitations of the available labels.

The precision values are especially low because precision was averaged across all 8,579 records, including records with no similar-patient labels. Those records counted as having no labeled matches. Recall was calculated only for the 4,802 records that had at least one similar patient listed. This difference in which records contribute to each average is important when comparing precision and recall. The overlap result provides another way to compare the methods: an average of 3.13 shared results means that the methods often chose the same patients, even though their evaluation scores differed slightly. These findings reflect the available labels and the methods used in this project, rather than a clinical assessment of whether a patient match would be useful.